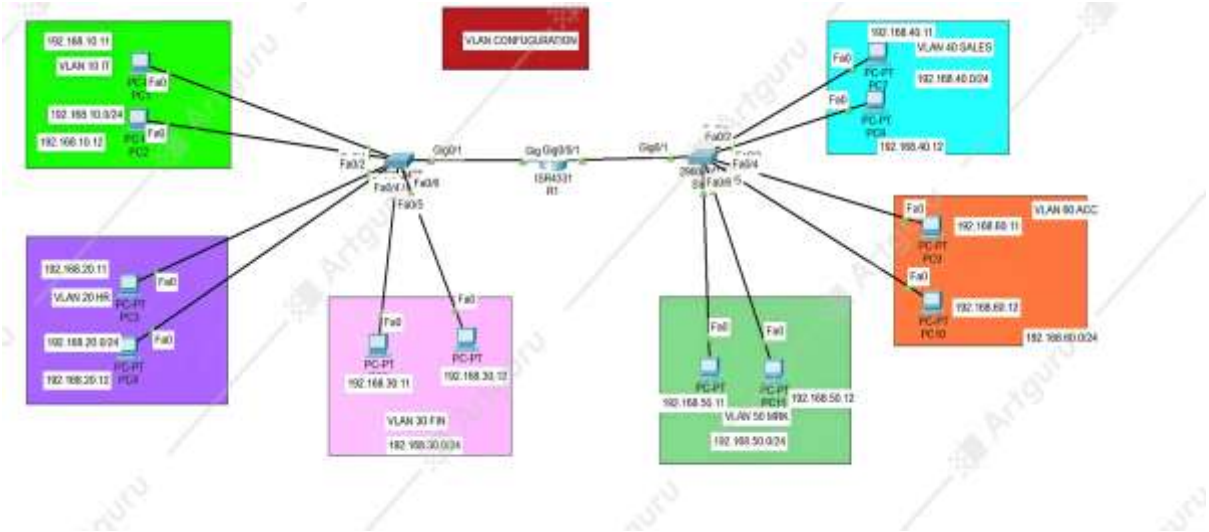
VLAN CONFIGURATION ON ROAS.



1. Hostname Configuration

Switch>enable
Switch#configure terminal
Switch(config)#Hostname SW1
SW1(config)#

Switch>enable
Switch#configure terminal
Switch(config)#Hostname SW2
SW2(config)#

Router>enable
Router#configure terminal
Router(config)#Hostname R1
R1(config)#

2. Enable Secret Configuration

R1(config)#enable Secret Asd123

SW1(config)#enable secret Asd123

SW2(config)#enable secret Asd123

3. Disable DNS Lookup

```
R1(config)#no ip domain-lookup

SW1(config)#no ip domain-lookup

SW2(config)#no ip domain-lookup
```

4. Creating VLANs

```
SW1(config)#vlan 10
SW1(config-vlan)#name IT
SW1(config-vlan)#exit
SW1(config)#vlan 20
SW1(config-vlan)#name HR
SW1(config-vlan)#vlan 30
SW1(config-vlan)#name FIN
SW1(config-vlan)#vlan 40
SW1(config-vlan)#name SALES
SW1(config-vlan)#vlan 50
SW1(config-vlan)#name MRK
SW1(config-vlan)#vlan 60
SW1(config-vlan)#name ACC
SW1(config-vlan)#exit

SW2(config)#vlan 10
SW2(config-vlan)#name IT
SW2(config-vlan)#vlan 20
SW2(config-vlan)#name HR
SW2(config-vlan)#vlan 30
SW2(config-vlan)#name FIN
SW2(config-vlan)#vlan 40
SW2(config-vlan)#name SALES
SW2(config-vlan)#vlan 50
SW2(config-vlan)#name MRK
SW2(config-vlan)#vlan 60
SW2(config-vlan)#name ACC
SW2(config-vlan)#exit
```

5. Assigning Access Ports

```
SW1(config)#interface range fastEthernet 0/1-2
SW1(config-if-range)#switchport mode access
SW1(config-if-range)#switchport access vlan 10
SW1(config-if-range)#
SW1(config-if-range)#interface range fastEthernet 0/3-4
SW1(config-if-range)#switchport mode access
SW1(config-if-range)#switchport access vlan 20
```

SW1(config-if-range)#
SW1(config-if-range)#interface range fastEthernet 0/5-6
SW1(config-if-range)#switchport mode access
SW1(config-if-range)#switchport access vlan 30
SW1(config-if-range)#exit

```
SW1#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/7, Fa0/8, Fa0/9, Fa0/10
                                                Fa0/11, Fa0/12, Fa0/13, Fa0/14
                                                Fa0/15, Fa0/16, Fa0/17, Fa0/18
                                                Fa0/19, Fa0/20, Fa0/21, Fa0/22
                                                Fa0/23, Fa0/24, Gig0/1, Gig0/2
10   IT                               active    Fa0/1, Fa0/2
20   HR                               active    Fa0/3, Fa0/4
30   FIN                              active    Fa0/5, Fa0/6
40   SALES                            active
50   MRK                              active
60   ACC                              active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
SW1#
```

SW2(config)#interface range fastEthernet 0/1-2
SW2(config-if-range)#switchport mode access
SW2(config-if-range)#switchport access vlan 40
SW2(config-if-range)#
SW2(config-if-range)#interface range fastEthernet 0/3-4
SW2(config-if-range)#switchport mode access
SW2(config-if-range)#switchport access vlan 60
SW2(config-if-range)#
SW2(config-if-range)#interface range fastEthernet 0/5-6
SW2(config-if-range)#switchport mode access
SW2(config-if-range)#switchport access vlan 50
SW2(config-if-range)#exit

```
SW2(config)#do sh vl br

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/7, Fa0/8, Fa0/9, Fa0/10
                                                Fa0/11, Fa0/12, Fa0/13, Fa0/14
                                                Fa0/15, Fa0/16, Fa0/17, Fa0/18
                                                Fa0/19, Fa0/20, Fa0/21, Fa0/22
                                                Fa0/23, Fa0/24, Gig0/1, Gig0/2
10   IT                               active
20   HR                               active
30   FIN                              active
40   SALES                            active    Fa0/1, Fa0/2
50   MRK                              active    Fa0/5, Fa0/6
60   ACC                              active    Fa0/3, Fa0/4
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
SW2(config)#
```

6. Configuring Trunk Ports

```
SW1(config)#interface gigabitEthernet 0/1
SW1(config-if)#switchport mode trunk
SW1(config-if)#no shutdown
SW1(config-if)#exit

SW2(config)#interface gigabitEthernet 0/1
SW2(config-if)#switchport mode trunk
SW2(config-if)#no shutdown
SW2(config-if)#exit
```

7. Configuring Inter-VLAN Routing (ROAS)

```
R1(config)#interface gigabitEthernet 0/0/0
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#interface gigabitEthernet 0/0/0.10
R1(config-subif)#encapsulation dot1Q 10
R1(config-subif)#ip address 192.168.10.1 255.255.255.0
R1(config-subif)#exit
R1(config)#interface gigabitEthernet 0/0/0.20
R1(config-subif)#encapsulation dot1Q 20
R1(config-subif)#ip address 192.168.20.1 255.255.255.0
R1(config-subif)#interface gigabitEthernet 0/0/0.30
R1(config-subif)#encapsulation dot1Q 30
R1(config-subif)#ip address 192.168.30.1 255.255.255.0

R1(config)#interface gigabitEthernet 0/0/1
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#interface gigabitEthernet 0/0/1.40
R1(config-subif)#encapsulation dot1Q 40
R1(config-subif)#ip address 192.168.40.1 255.255.255.0
R1(config-subif)#
R1(config-subif)#interface gigabitEthernet 0/0/1.60
R1(config-subif)#encapsulation dot1Q 60
R1(config-subif)#ip address 192.168.60.1 255.255.255.0
R1(config-subif)#
R1(config-subif)#interface gigabitEthernet 0/0/1.50
R1(config-subif)#encapsulation dot1Q 50
R1(config-subif)#ip address 192.168.50.1 255.255.255.0
R1(config-subif)#exit
```

```
R1(config)#do sh ip int br
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0/0   unassigned      YES unset  up                    up
GigabitEthernet0/0/0.10192.168.10.1    YES manual up                    up
GigabitEthernet0/0/0.20192.168.20.1    YES manual up                    up
GigabitEthernet0/0/0.30192.168.30.1    YES manual up                    up
GigabitEthernet0/0/1   unassigned      YES unset  up                    up
GigabitEthernet0/0/1.40192.168.40.1    YES manual up                    up
GigabitEthernet0/0/1.50192.168.50.1    YES manual up                    up
GigabitEthernet0/0/1.60192.168.60.1    YES manual up                    up
GigabitEthernet0/0/2   unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
R1(config)#
```

8. Static IP Configuration for all PCs

9. Connectivity Testing